Real people in surreal environments

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With high performance computer processing capabilities and appropriate communication bandwidths, virtual reality (VR) and enhanced reality systems are no longer restricted to sophisticated flight simulators. Now there is the opportunity for users to immerse themselves in new worlds and to learn, work and play in different ways. This paper presents some of the key research issues that need to be addressed in order to develop virtual environments (VEs) which will 'succeed' across a range of VR application areas, including: What are the key components that constitute compelling, stimulating and easy-to-use VEs? What constitutes a usable VE? Do we understand the human element of VR? Techniques and approaches that will increase our understanding of VR from the user perspective are presented, together with 'The Mirror' case study which demonstrates the assessment methodology used to address performance and usability issues relating to a collaborative environment delivered over the Internet.

1. Introduction

> "It is not the technology ... what matters is what happens in the hearts and minds of the users."
>
> (Judith Robin, Worlds Away, E2A '96)

The emergence of virtual reality (VR) is dramatically changing the way we communicate. It is redefining the role of computers and telephony to provide compelling virtual environments (VEs) that help mediate collaboration, conversation and social interaction between people.

Advances in technology over the past decade have led to geographical boundaries coming down as we start moving towards an age of networked communities. The concepts of telepresence, teleoperation, virtual conferencing and shared environments are becoming a reality in supporting these communities.

Preliminary research has generated a wealth of information and insight into this area [1—3]. However, it is realised that as application areas for VR become more refined and the choice for the user increases, it will be the provision of intuitive, compelling, and easy-to-use interfaces that will become the key differentiator. This paper looks at some of the issues surrounding these new ways of communicating. An overview is provided of the human perception of the physical aspects of VR, highlighting the need to understand human perception and the interaction between the senses, and describing the human cognitive element in VR, the approaches and the techniques used for designing and developing compelling and usable VR applications.

Understanding of the user issues related to VEs delivered over the Internet has increased significantly over the last year through field trials and concept testing, and specific areas for further research have been identified. Researching the human element of collaborative VEs is still at an early stage. The aim is to understand and address these issues across the different media spaces available for delivering VR.

In section 4, it is stressed how user perception of a VE consists of complex interaction between the various elements which contribute to a successful interaction.

Finally, an example of an Internet VE, The Mirror [4], is provided to demonstrate the assessment methodology used to address technology and the interface factors required for delivering successful VEs.

2. Physical aspects of virtual environments

Virtual reality can be applied to a wide range of application domains including training, business, simulation, education, design, manufacturing, medicine, and entertainment.

2.1 Presence

For there to be benefit and added value over conventional means of communicating, interacting within virtual environments must be compelling, efficient, intuitive and responsive. However, virtual environments offer more than images and sound, they can offer presence (or, synonymously, telepresence, in the case of distributed or shared environments).

Presence is not new and can mean different things to different people. For a long time playwrights, artists, film and TV producers have used devices to engage audiences through their particular medium. Presence is hard to define and even harder to measure. However, most definitions of presence include the notion of 'being there' and the challenge is to engineer systems and devices to fool the senses into maximising this notion and thus produce systems that are sufficiently immersive for the application. The range of specification or granularity for virtual environments spans a wide range, including:

- high-end flight simulators incorporating high specification visual, auditory, motion and tactile display and feedback,
- head-mounted stereoscopic displays with motion tracking,
- SmartSpace [5] with a widescreen display and acoustic bubble,
- PC or workstation with a physically narrow field of view compensated by appropriate content, e.g. net-worked games can offer some of the criteria described above using quite modest resources.

One of the principal ingredients that impacts on this sense of presence is the ability to perceive spatial information enabling navigation and interaction with virtual surroundings. The essential location information is in the form of the depth, volume and space cues which can be provided by a number of possible techniques including:

- occlusion or interposition whereby more distant objects can be hidden by nearer objects,
- stereopsis as a result of binocular disparity which is particularly significant when 3-D objects are close to a viewer,
- perspective,
- the use of shadow and motion parallax.

These visual cues can also be complemented by audio cues associated with the visual elements such as Doppler shift or appropriate reverberation profiles.

2.2 The visual environment

In a virtual environment the visual information covers a range of complexity:

- simple wire-frame graphics,
- solid objects,
- rendered objects,
- photo-realistic models.

There might be a temptation to assume that high levels of graphical detail in VR environments is desirable or essential. However, there are times when scenes can become too intricate or cluttered resulting in the key elements being less effective. At times, more is better, less is more.

2.3 The audio environment

The audio environment is often underestimated and can be exploited to good effect for most applications if there is:

- adequate scope for accurate manipulation and synthesis of the soundfield,
- good temporal and spatial correlation, such as synchronisation, between the audio and visual geometric cues or video components.

The general audio quality in terms of level, noise and distortion performance should be commensurate with any corresponding video quality. However, when virtual environments are shared with other people and conversational speech is a major component then it is also necessary to ensure that there is adequate echo cancellation and minimum transmission delay. Likewise, if voice switching is employed then there should be good immunity to any extraneous background noise and it should not clip the speech to any significant degree.

2.4 Contrasting the contribution of audio and video

There can be a danger in considering audio and video perception as separate entities. The relative significance or importance of the audio and visual channels are factors that need to be carefully considered to gain the optimum effect.

Evidence of subtle cross-modal interaction between these channels has been demonstrated in a number of cases, for example:

- perception of TV picture quality is affected by the presence of audio [6],
- the McGurk effect [7] illustrates how the ears can be tricked into hearing different sounds when the eyes are open than when they are closed,
- a highly detailed, photo-realistic, VR fly-through of a building will have a 'dead' feel to it if just the visual channel is used — the addition of even low-level acoustic background sounds or incidental noises can heighten the feeling of being part of the environment being visualised and contribute to our sense of presence.

The audio and video components of a virtual environment have fundamentally different properties, as different views on to a common virtual world can be seen on individuals' displays at the same time and in the same room with negligible interference. This is not the case with audio as sources from several directions, including behind the listener, can be heard simultaneously. As a result, consideration needs to be given to locations where different virtual environments are physically close to each other in order to minimise any acoustic interference between them.

There is a further potential enhancement that can be gained from the inclusion of an appropriate audio environment. In some applications it is possible to maintain a degree of continuity of presence so that when the head is turned away from a display system the audio can still be heard. If, however, the user was restricted to a visual stimulus and turned away, then any sense of presence that was being experienced would be lost.

2.5 Network considerations

Many interactive VR environments have been developed for the simulation and CAD (computer aided design) industries; these are generally implemented on local high-performance workstations, involve a single user and are normally limited by the local hardware and software. When virtual environments are to be shared among users who are not collocated, the performance of different network technologies and topologies needs to be considered.

Delivery of interactive VEs can span from public Internet connection with bandwidths constrained by comparatively low-speed modems and copper cables, through to high-speed wide-bandwidth photonic networks.

Apart from the fundamental limitations of bandwidth that might impact on audio quality and video information, transmission delay is a major consideration when real-time responses and interactions are anticipated.

3. Researching the human element

> "You [the user] not only give your avatar [1] life, but you give other avatars life via your presence. Pixels do not matter unless you have people behind them."
>
> (Randy Farmer, Electric Communities, E2A '96)

> "It seems like the people really make the worlds."
>
> (User comment from The Mirror trial)

People are the main content of CVE (collaborative virtual environments). It is through their actions and experiences that the environment comes to life. Technology is the enabler to achieving user goals and tasks effectively and efficiently. As a result, technology must be harnessed to support the user experience and must not be the sole driver.

With collaborative VEs, we are moving into a new domain where the user becomes an integral part of the technology. Researching the human element requires a modification of more traditional approaches in order to cater for the fact that, unlike most other technologies, with VR we are dealing with a dynamic media where the environment is generated in real time in response to user actions and experiences. User experiences vary depending on how they decide to interact with objects in the VE and also who is sharing the space at the time.

There is a tendency for HCI (human/computer interaction) to reuse and reinvent techniques drawn from its own narrow sphere. Successful design, development and evaluation depends on collecting approaches from different domains and tailoring these to the different life-style stages in the product development life cycle. To succeed, novel approaches need to be developed by combining techniques such as storyboarding and ethnography with existing traditionally used ones such as user trials. In addition, multidisciplinary teams are essential for the development of successful VR applications, including a combination of 3-D designers, software designers, human factors specialists and service providers.

Below are some examples of approaches to increase understanding when designing for the user.

- Requirements capture through visualisations

  Techniques using early visualisations of complex concepts have been developed which can be taken to users early on in the development phase. This was applied to the information place (IP) [1], a 3-D application involving the use of an astral metaphor for

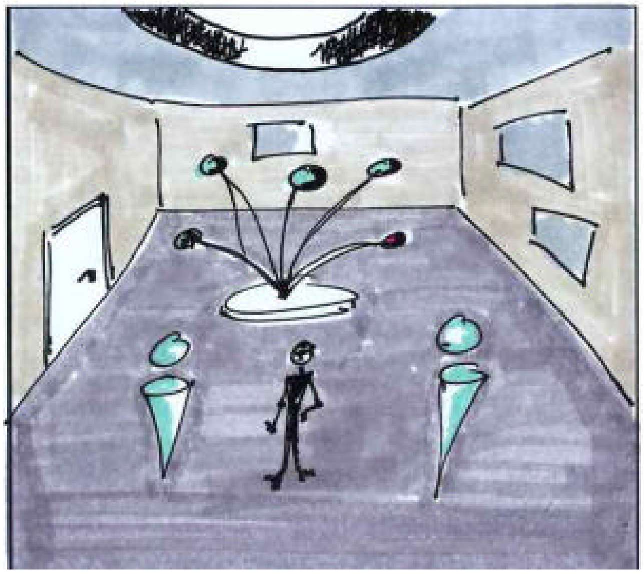
[1] Individuals are represented in virtual environments through their avatars — a body image generated by the computer and controlled by the user.

organising and representing information in a collaborative virtual environment. At a top level, the user sees a constellation of planets which represents the different groupings of information. The user can navigate around the space visiting planets, meeting people and interacting with objects and information.

The information place is an example where technology and design principles were used to create an innovative demonstrator. Early visualisations, in the form of story-boards were taken to a potential segment of users (business users) to gauge their requirements for such a virtual environment.

Again, the requirements focused on the key components of CVEs, such as navigation, avatar representation, and communication. Additional issues, including security, business versus personal use, were also addressed.

The use of storyboards as a technique for visualising concepts proved very effective in demonstrating the concept and eliciting key requirements for the IP. Figure 1 shows an example of a clip from the IP storyboard. The image shows the astral metaphor and the scenario sets the scene.



When Rob has finished he notices Jane's avatar. Jane has been discussing her work with Peter and when they see Rob enter they invite him to join the conversation .....

Fig 1 Sample from the information place storyboard.

This technique can be implemented cost effectively and very quickly with maximum impact. Feedback from such exercises can be used to implement a range of potential prototypes based on an element of user attitude and innovative design. These prototypes can then be tested by potential users to further refine understanding from the user perspective.

Other techniques that can be used to capture requirements include the use of rich pictures to obtain a diagrammatical representation of the users' thoughts through their own drawings, and careful observation of the real-world situations which are to be supported in the VE.

- Understanding VR through users' actions

The walk-through technique has been used to understand:

— the usability of particular features from the user perspective,

— thought processes in relation to problem solving and task interpretation,

— how users deal with errors encountered when using an interface.

Based on the 'think aloud' technique, users speak out their thoughts and actions, describing what they like and dislike about the particular interface they are using. This approach can be used in combination with other techniques, such as part of the requirements capture process, in user trials. This strategy was used early on in VR research to identify components and issues that are important to users and which contribute to 'usable' virtual environments. The data was used to inform designers and contribute to the development of a questionnaire for assessing VEs.

- Controlled user trials

User trials allow researchers to assess user attitude and performance in a controlled environment. Tasks can be designed to assess particular aspects of the system under trial. For example, in The Mirror beta trial, users were asked to complete a set task in one of the worlds where they had to collaborate with other avatars to complete the task.

Depending on the design of the user trials, they can be used to compare different implementations, e.g. a comparison of immersive environments with non-immersive environments to investigate ease of use [8].

- Field trials

The key advantage a field trial has over a user trial is that it allows researchers to investigate usage over time and in the real context of use. In the absence of controlled tasks, field trials of virtual environments pose new challenges for HF researchers. With the main Mirror trial (see section 5), there was very little opportunity to control who went into the world, and at what times. As a result a range of data-gathering techniques, existing and new, were employed to collate objective and subjective data.

Additional approaches to assessing VEs currently being considered include:

- ethnography [9], an observational technique which can be used to analyse user interaction/communication within real environments to establish salient attributes that can be supported within VEs,
- modelling techniques to understand behavioural changes and social implications of CVEs,
- spatial syntax theory [10], to understand the relationship between space, movement and interaction in VEs.

4. What contributes towards a usable virtual environment?

The scope of research required in this area is vast and some of the questions that need to be addressed include the following.

- Which tasks are most suitable for users to perform in VEs?
- What are the physiological effects on the users? Will users get sick or be adversely affected by exposure to VEs? How much sensory feedback from the system can the user process?
- How can VE technology be improved to better meet the user's needs?
- Do design metaphors enhance a user's performance in VEs?

Over the past year BT has undertaken preliminary research to address the design, technology and human factors issues involved in delivering virtual environments over the Internet. The objectives of the research were to assess the technical performance of a CVE, and to investigate user perceptions related to inhabiting 3-D virtual environments.

An early study in BT consisted of a series of user walk-throughs carried out with novice users. The study addressed the fundamental goals of interacting with virtual worlds, e.g. communication and navigation, in order to establish issues that are important to potential users.

The main aims of the study were to assess the usability of particular components that make up a virtual world from the user perspective, and to identify particular aspects of the worlds that users like and dislike. People were asked to use three virtual worlds which were accessible in real time over the Internet. They were instructed to interact with each world in order to get an overall feel for what each had to offer. As a secondary task, they were asked to communicate with other avatars in the space.

The study identified the following components that contribute to the users' experience of a VE available over the Internet:

- entry into the VE, and initial impressions,
- communication and collaboration in virtual environments,
- avatar representation, customisation and animation,
- navigation,
- overall look of the world (i.e. the use of graphics),
- use of audio/visual cues,
- sense of presence.

The above is a top-level list which can be broken down further. For example, avatar representation includes methods for selecting and changing appearance, facial/body movement, expression of emotions, how to present the information to users, etc.

Some of the main conclusions from the study are now presented. There is no one component that makes a successful virtual world. Navigation, communication, and system feedback are all intrinsically related. This was captured in a user's comment:

> "They are sort of equally important. Being able to choose your avatar is very important. Conscious of how you look and how others are perceiving you. At the same time, what's the point of being able to navigate wonderfully if you can't communicate with somebody. What's the point of communicating with somebody if you can't see them; you might as well be on the phone. You kind of need it all."
>
> User comment

The goal is to understand the relationship between the three different components.

4.1 Communication

Communication was seen as the key component within any CVE. Users need to communicate and interact with other people, information and application objects. All other components contribute towards making communication easier for the user — navigation, avatar representation, choice of text or speech, use of sound and motion, and the general look of a world.

The study highlighted some of the key steps for supporting communication within the CVE.

- Making the decision to communicate

  The virtual environment must support users in their decision to communicate; this decision could result from an interest in the appearance of another avatar, a particular motion indicating interest, information held on avatar profiles, or hearing other conversations. Such information must be both easy to convey and access.

- Deciding on the type of communication

  Users may require a range of communication 'types', e.g. one-to- one, one-to-many, group chat or formal meeting, public chat, private chat as part of a larger group chat, etc. Again these should be supported within the shared environment.

- Initiating the communication

  There are numerous techniques used to initiate communication in the real world. These range from a simple informal one-to-one chat resulting from chance eye contact, through to a pre-arranged formal meeting set up in advance over the telephone, by e-mail or by a third person. Virtual worlds need to provide the necessary tools to support such requirements, e.g. the use of gestures or specific movement can be used to attract someone's attention. A noticeboard or an on-line e-mail facility can be used to set up a larger chat.

- Carrying out the communication

  The virtual world needs to support user requirements during the communication, e.g. use of gestures (facial expressions and body movement, availability of people through their representations), use of position and location in the space to provide feedback cues to other avatars, easy access to information, and availability of virtual equipment, such as whiteboard, document transfer facility, the ability to invite others to join a chat, etc.

The above is by no means an exhaustive list and the requirements will vary depending on the user population.

Communication will be greatly influenced by the use of text or speech, each of which has its good and bad aspects (see Table 1).

Table 1 Advantages and disadvantages of speech and text

Method of communication	Advantages	Disadvantages
Text	Gives users time to compose messages.	Open public chats using the same window can be potentially confusing, as a number of separate conversations are presented which can increase the amount of processing required by user.
	Users have a record of past conversations.	May detract from realism. If text is in a separate window, it can draw attention from main window.
	Helps communication if text can be easily attributed to avatars within the environment.	Typing introduces significant delays into the communic-ation process.
Speech	Users perceive speech as the natural form of communication between people.	Technical constraints can lead to distortion.
	Has the potential of adding to the sense of presence.	Potentially confusing if all conversations are broadcast.

The advantages and disadvantages are dependent on the system implementations. A combination of technical advances and design can help overcome the current limitations.

Other issues that impact on ease of communication include the speed of the system, implementation of the aura, and use of audio and visual stimuli. Each needs to be implemented differently depending on the specific application. For example, for entertainment, it may be necessary to go against user expectations, whereas for business use, it is more important to map user expectations.

4.2 Navigation

Navigation impacts on a user's perception of the quality of the interaction and sense of presence. As such, navigating in a virtual environment should be as intuitive as possible requiring minimum effort from the user. Users need to be able to navigate easily so that moving around in the VE does not detract from the main goal of communicating.

The key elements that contribute towards 'intuitive' navigation include the response time, contextual information and system feedback in terms of the visual information. This becomes more important in a text-based world where the text window is separate from the main graphics window. Without feedback the graphics window becomes redundant as all the 'action' is in the text window.

Different implementations of navigation within VEs are currently being investigated to reduce the amount of effort required on behalf of the user.

4.3 Use of audio

Audio plays an important role by helping to maintain continuity in the VE. It enhances the sense of presence, providing users with realistic stimuli in the form of ambient cues and collision cues. Not only does this add to the sense of presence, but it also aids navigation and communication.

In summary, all these aspects of CVEs need to be implemented to reduce the load on the user. They need to become part of the automatic background functionality.

5. A method for assessing virtual environments — The Mirror case study

> "A rapidly expanding system of networks, collectively known as the Internet, links millions of people in new spaces that are changing the way we think, the nature of our sexuality, the form of our communities, our very identities."
>
> (Sherry Turkle, *'Life in the Screen'*, 1995)

In addition to researching specific aspects of VEs, it is essential for developers and designers to learn about the whole process of delivering virtual environments to end users — the customer experience. This involves consideration of the technology, system performance, and content design, together with the user interface issues. The Mirror project provided an ideal opportunity to investigate these areas.

The Mirror was a research project created by BT Laboratories, Sony Corporation, the BBC and Illuminations. The project brought together key disciplines of 3-D design, media, engineering and human factors to produce six compelling and stimulating 3-D virtual interactive worlds. For a detailed description of The Mirror, see Walker [4].

The worlds in The Mirror reflected the themes of the six broadcast programs on the BBC2 technology series, The Net. They were designed to experiment with specific aspects of VR content, to explore what would be most appealing to both new and experienced participants. The worlds were accessed via the entry portal (Fig 2) which highlighted the 'World of the Week' corresponding with the programme.

When designing the world, one of the goals was to develop a self-sustaining environment in which users contribute to the content of the VE not only through communication, but also through involvement in the special events, providing material for the art show in the Creation world, leaving memories in Memory world, and interacting with the application objects within the worlds.

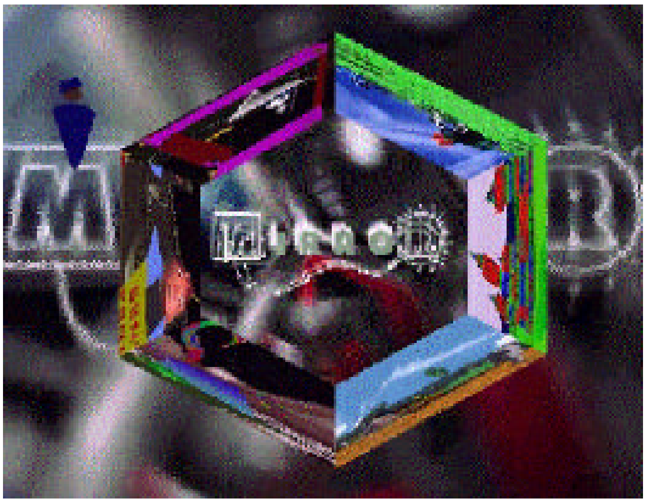


Fig 2 Entry portal for The Mirror.

The assessment of The Mirror, which consisted of two phases, looked at the success of these features.

- Beta usability trial

  This was a controlled user trial which ran over a two-week period. The overall aim was to test the process for designing and delivering a virtual environment to end users in preparation for the 'live' trial with the BBC. The main trial consisted of four one-hour sessions in which the participants were asked to complete a task.

  Additional assessments were carried out to evaluate the Sony browser usability, prepare the guidance material for the main trial and design the cover for the CD.

- The Mirror field trial

  The main trial ran for 7 weeks between 13 January and 28 February 1997. The main aim was to investigate and understand the technology and interface issues required for delivering interactive 3-D environments over the Internet. This extended trial allowed for changes in usage to be noted over time. It was also possible to find out what users had expected from the technology.

5.1 Measuring usage and usability in The Mirror CVE

It is believed that a comparison of actual (objective) behaviour and perceived (subjective) feedback is required to provide an overall assessment of usage and usability. Subjective assessment examines user attitude while objective assessment involves logging platform statistics to establish actual usage and behaviour.

Objective analysis (quantitative)

Usage data, which took the form of performance statistics of interaction between users and the VE, included individual data and average data for all users. Types of measurements included frequency of access and total time spent in each session, etc.

Recorded interactions between users and the VE included the analysis of the text files of communication between users. Where possible, recordings of visual content were also used.

Subjective analysis (qualitative)

On-line questionnaires/focus groups were used to target a sample of users to obtain feedback related to all aspects of the trial. Focus groups were carried out both face to face and in the actual Mirror VE.

Evaluators accessed and observed behaviour within the worlds, both as passive and active participants.

Helpline data included a breakdown of problems and issues raised via the 'Helpline'. These contained both technical and usability-related issues.

All correspondence between the users and the assessment organisers was carried out via the Internet.

5.2 Results from The Mirror experiment

Some of the high-level results are presented below:

- over 2300 people registered to become citizens of The Mirror spending a total of 4408 hours logged on to the server (see Fig 3),
- in terms of the content of the worlds, the graphics and the interactive applications provided an initial focus of interest which was essential in attracting people and ensuring return visits,
- the special events played a key role in bringing together new people to share novel experiences with attendance totalling over 500 hours across all the worlds — such events contributed to an evolving 3-D space where users' experiences and interactions change the environment,
- users who became regular visitors to The Mirror, developed a real sense of community over the trial period and feedback indicated a genuine emotional involvement rarely associated with new technologies,

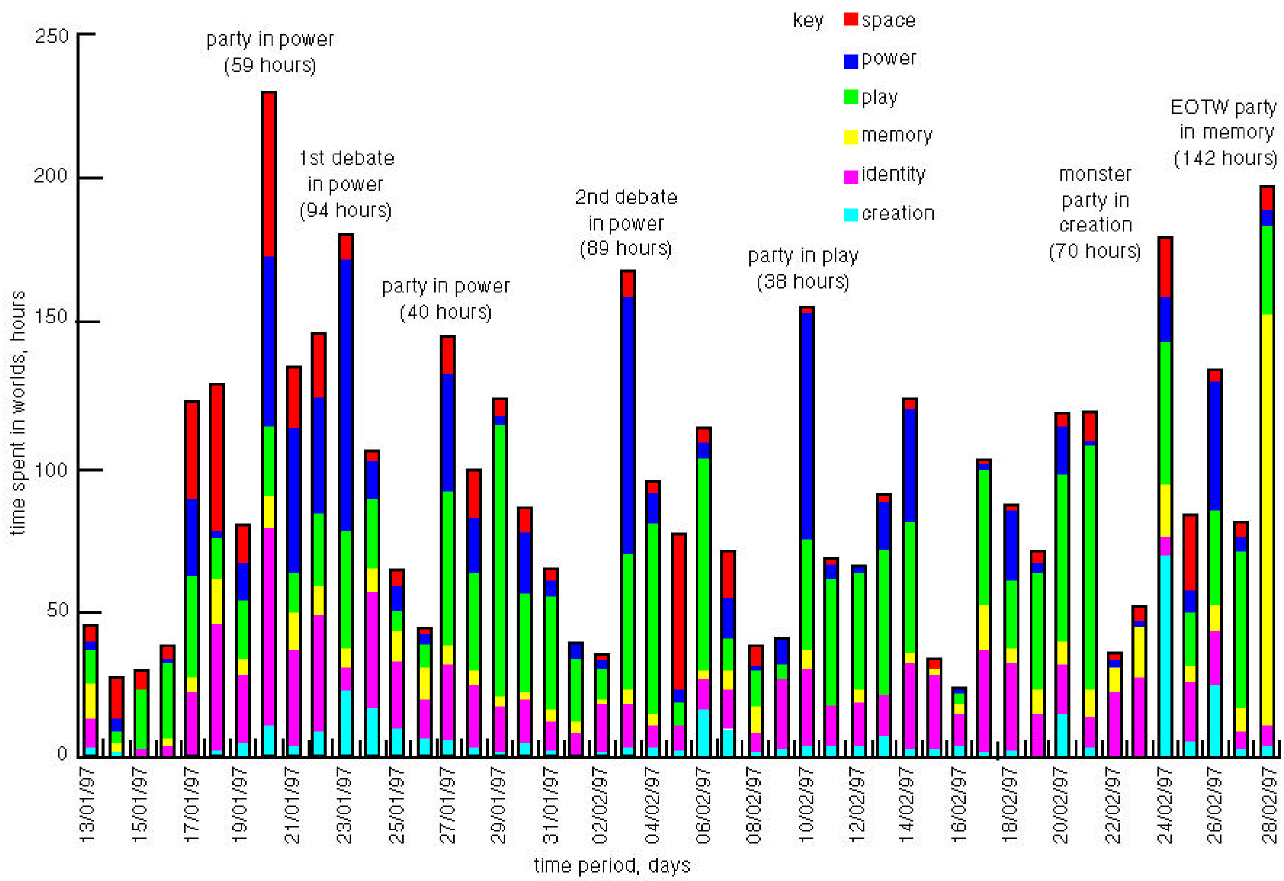


Fig 3 User hours spend in The Mirror.

- the main reasons for return visits were to meet new people, look around in the worlds, and interact with objects (in that order) — over time, communication between users was the most important reason,

- avatars were perceived as essential in allowing individuals to project their own personalities — The Mirror included a choice of four avatars which participants could customise (see Fig 4), with users highlighting the need to include additional functionality, e.g. gestures, use of emotions, functionality to enhance self-awareness in the VE.

Fig 4 The Mirror avatar changing room.

The trial highlighted four main critical issues:

- initial contact with The Mirror — the early contact with customers (the initial advertising, registration process, packaging, and delivery of CD and installation process) are key to providing a positive first impression,

- speed of loading worlds/interaction speed — long loading/response times resulted in some users abandoning the trial at the onset,

- the existence of a personal aura was seen as a major barrier to communication — for small numbers of users, this does not cause too much confusion, but when large numbers are gathered in a small space, restrictions arising from the aura mean that the users may not be able to see other people they are expecting to see, e.g. if the aura size is ten, it may be that the eleventh person expecting to meet friends may seem to be alone or with complete strangers,

- achieving a critical mass of people within the worlds — the primary motivator for visiting or returning to a particular world was the number of people in it, with users reporting that many worlds were empty, thus reducing motivation to return there in future.

6. Discussion — future direction

This paper has provided an overview of some of the factors that will influence users' decisions both to exploit and, in the longer term, to inhabit virtual media environments.

In order to develop VEs which are effective and well received by their users, the following areas have to be better understood:

- perceptual aspects of VEs, e.g. an understanding of human perception and the interaction between the senses,

- design of the content for compelling, persistent VEs,

- usability of CVEs, e.g. impact of navigation, sense of presence, communication, etc, on the user experience,

- physiological impacts of VEs on the user,

- social/ethical implications of VE technology,

- platform performance threshold required to deliver successful VEs across different media environments.

Clearly, there is a diverse range of possible applications and implementations of virtual environments and as a result there is a need for developers to be able to measure, benchmark, quantify and model systems. The basic properties of hearing and vision are fairly well understood and there are evolving recommendations from bodies such as the ITU for the assessment of audio and visual quality. However, when users are immersed in a virtual environment there is a resultant notion of presence setting new challenges requiring new metrics and assessment techniques.

In addition to understanding the perceptual issues associated with physical VE interfaces, the techniques for the capture of user requirements, design and implementation of key functions such as communication, navigation and the selection and manipulation of avatars are being developed to provide appropriate and effective implementations. The HCI field has developed models, techniques and methodologies for understanding the 'human element' across different products and services. However, with virtual reality there is a need to revisit the way in which these techniques are used.

Unlike many other available technologies, virtual reality is a totally new type of interface which is dynamically changing and evolving in response to the actions of users. Researchers and developers need to understand the nature of this medium, both from a technical and user-centred point of view. The challenge is to develop novel ways of combining and using new and existing tools and techniques that will help to effectively understand, design, develop and evaluate this fast-moving medium.

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She is currently leading the human factors team on the 'understanding virtual environments' (UVE) task, and was responsible for the usability assessment of 'The Mirror' and 'Heaven and Hell' shared spaces experiments.

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